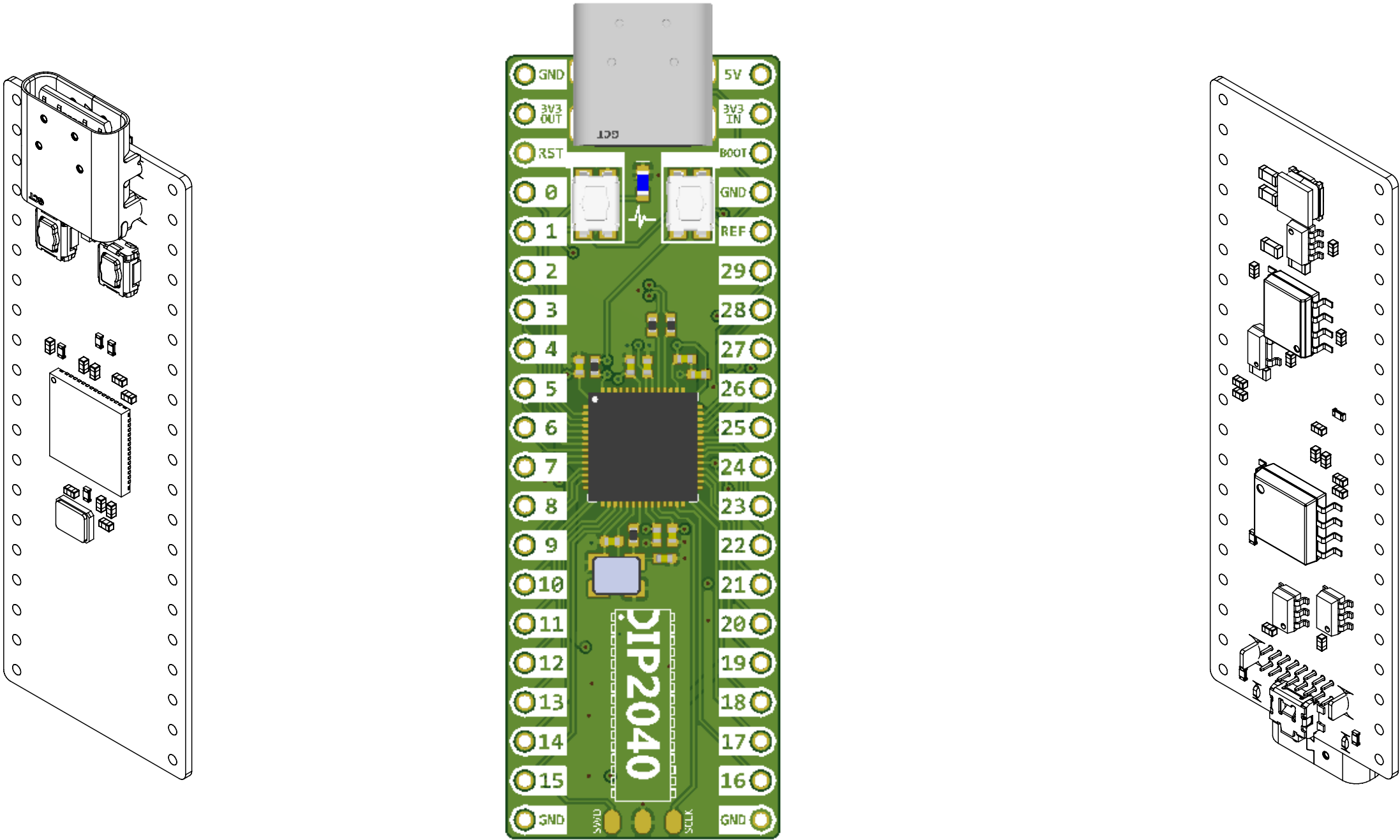


DIP2040.PrjPcb

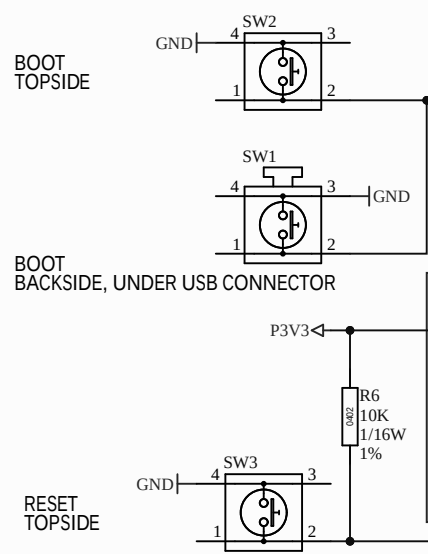
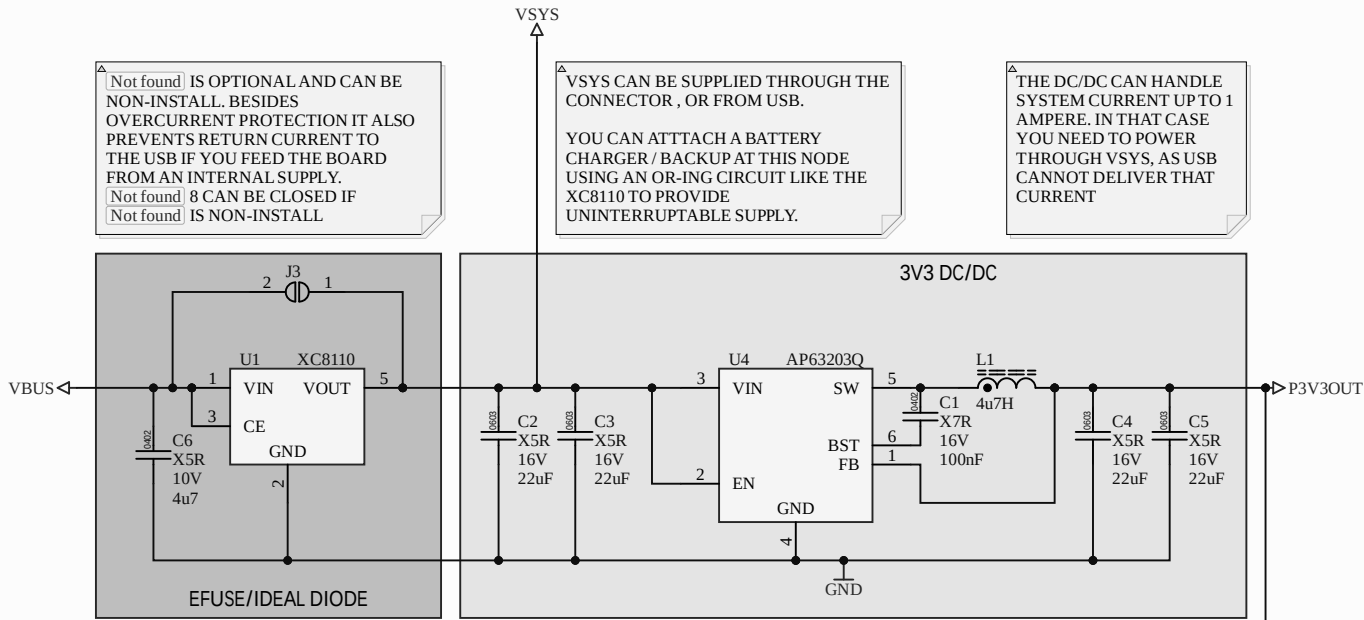
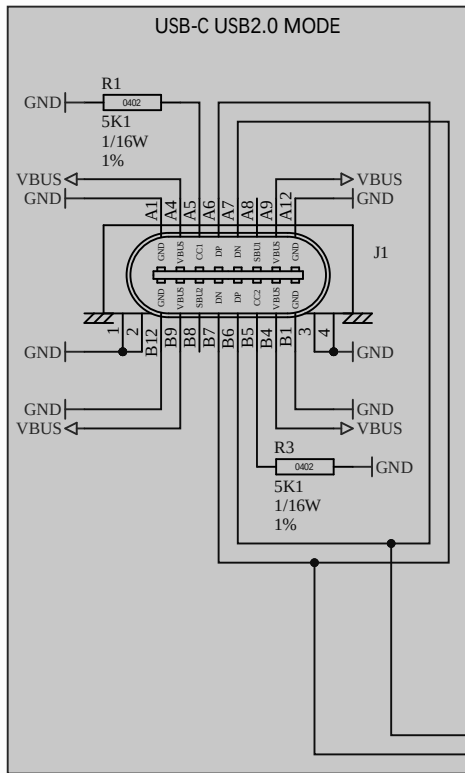


Document Creation Date: 7/4/2025

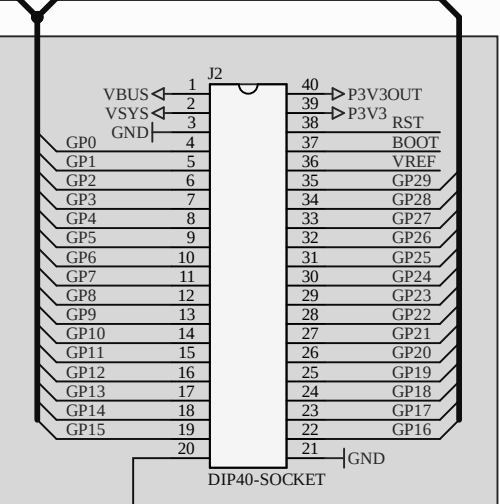
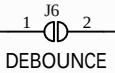
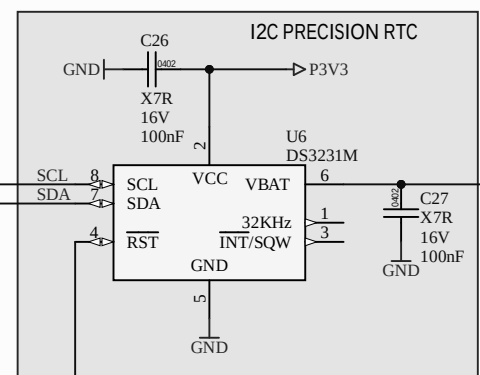
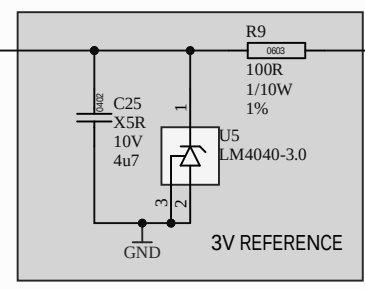
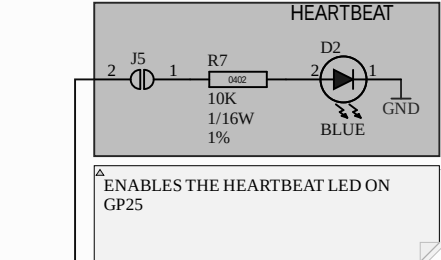
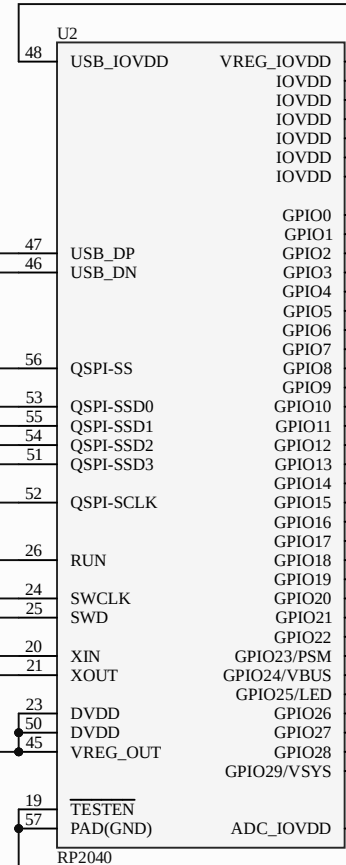
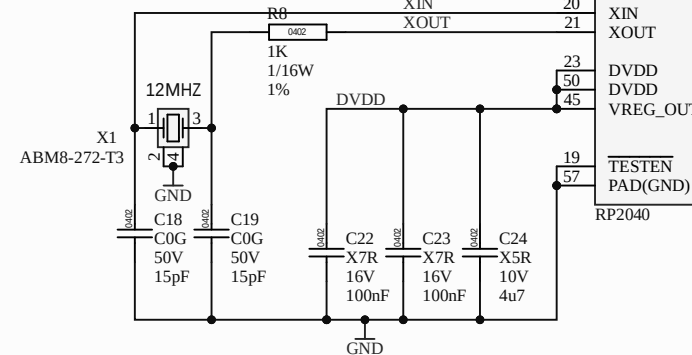
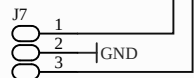
Design : Vincent Himpe



DIP2040
RP2040 BOARD



SW1 IS MOUNTED ON THE BACKSIDE, UNDER THE USB CONNECTOR SO THE CPU CAN BE RESET USING A PAPERCLIP THROUGH AN ENCLOSURE



1	2	3	4	5	6
GENERAL					
A	GENERAL				
	1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA				
	2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED.				
	3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER				
	4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2				
	5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED				
B	6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS				
	7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT.				
	8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER				
	9. SURFACE FINISH : HASL				
	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
			Fullyrouted Project DIP2040.PrjPcb Version: Variant [No Variations] FABRICATION DRAWING		
1	2	3	4	5	6

LAYER STACK

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Material Type	Gerber	Dk	Weight	Constructions	Df	Resin
		Top Overlay			Legend	GTO					
	Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
	Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
	Prepreg		0.071mm(2.800mil)	PP-006	Dielectric		4.1		1080	0.02	62%
	CF-004	Layer 1	0.035mm(1.378mil)		Signal	G1		1oz			
											
			0.320mm(12.600mil)	FR-4	Dielectric		4.8				
	CF-004	Layer 2	0.035mm(1.378mil)		Signal	G2		1oz			
	Prepreg		0.071mm(2.800mil)	PP-006	Dielectric		4.1		1080	0.02	62%
	Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
	Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
		Bottom Overlay			Legend	GBO					
Total thickness: 0.624mm(24.556mil)											

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
○	64	0.254mm(10.000mil)	Plated	Round	Top Layer - Bottom Layer	Via					
✕	16	0.400mm(15.748mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
□	40	0.800mm(31.496mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
▽	2	1.400mm(55.118mil)	Plated	Slot	Top Layer - Bottom Layer	Pad	Rounded Rectangular				
⊗	2	2.100mm(82.677mil)	Plated	Slot	Top Layer - Bottom Layer	Pad	Rounded Rectangular				
	124 Total										

DRILL DRAWING

A

A

B

B

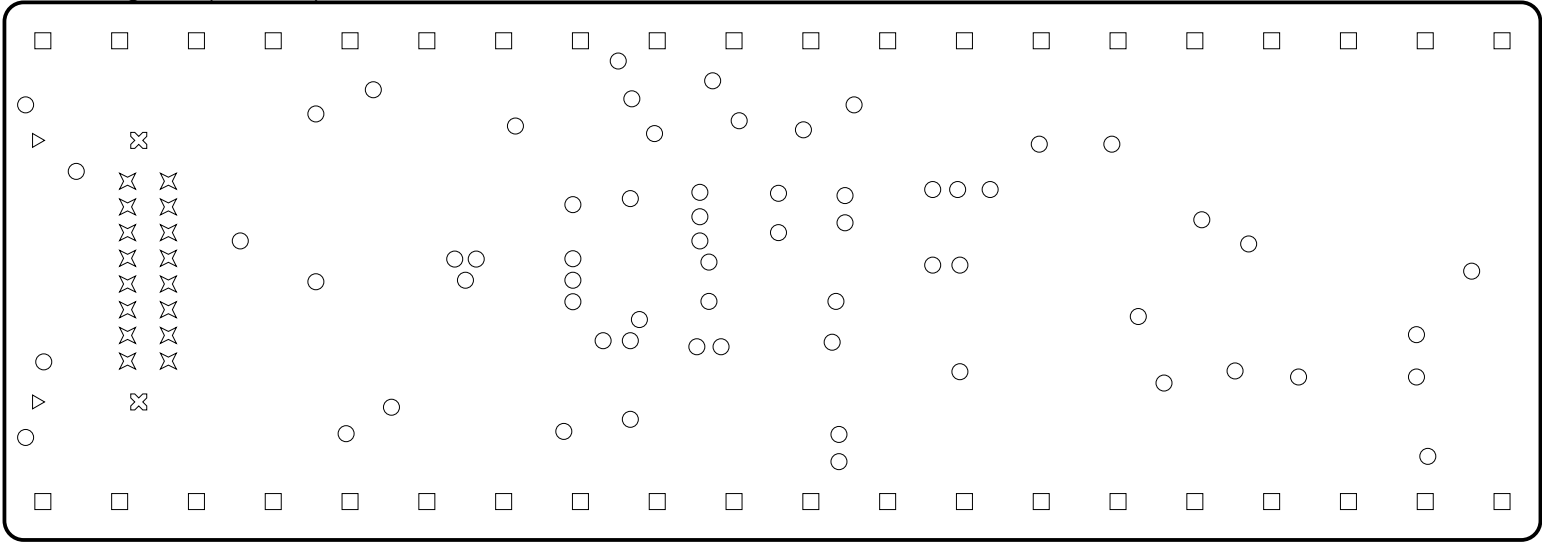
C

C

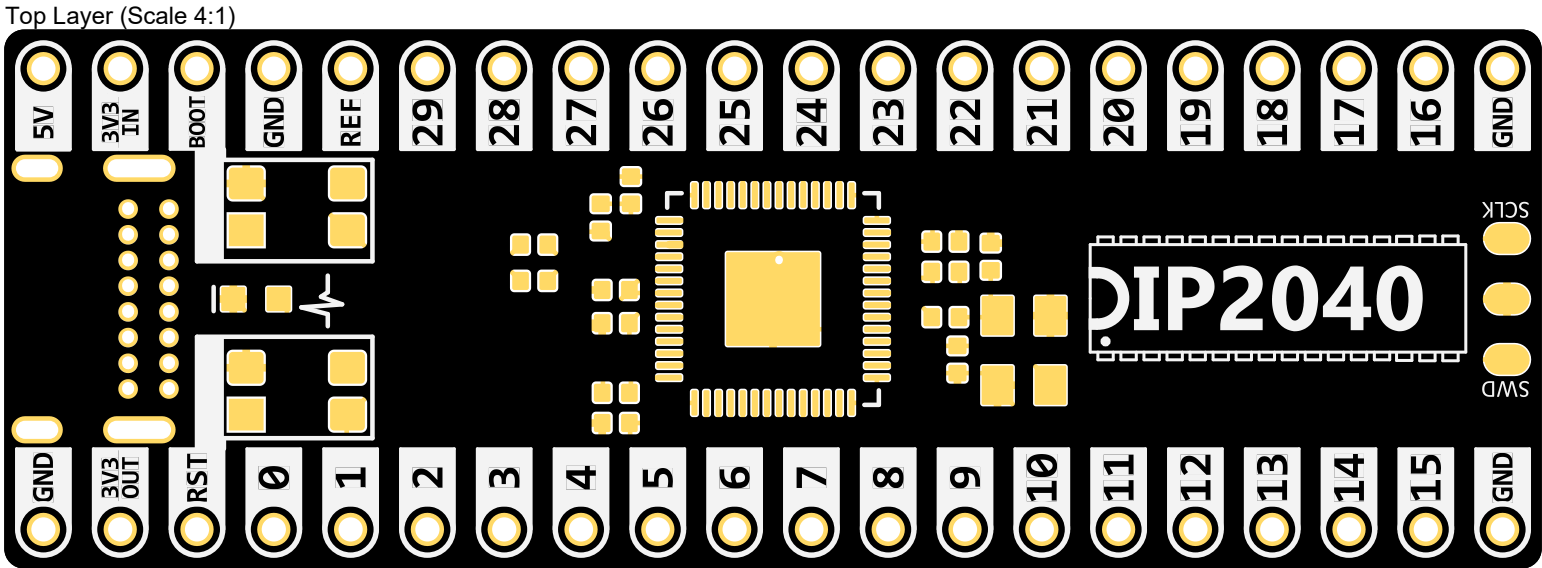
D

D

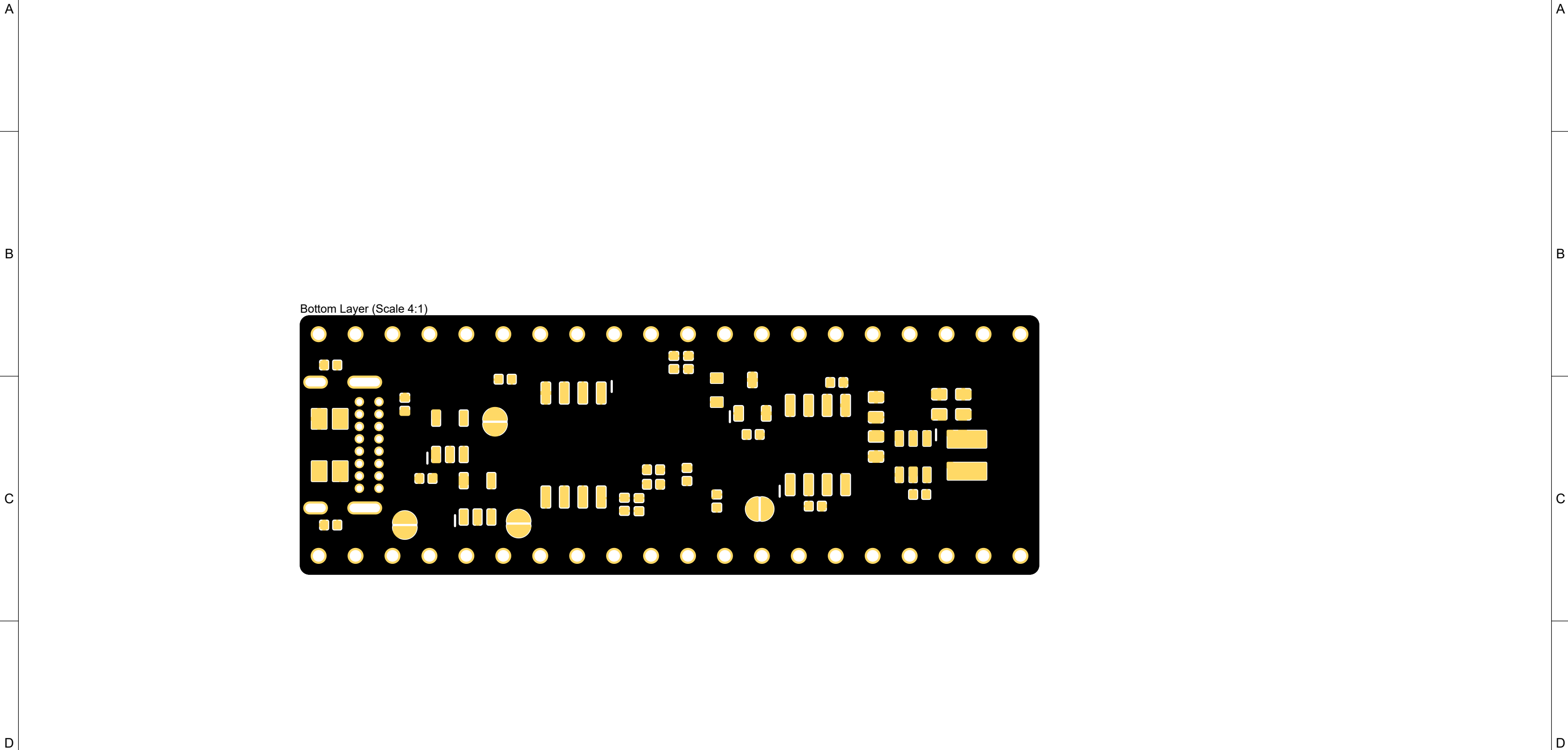
Drill Drawing View (Scale 4:1)



COMPOSITE VIEW FRONT



COMPOSITE VIEW BACK



37

38

39

40

41

42

LAYER VIEW : TOP LAYER

A

A

B

B

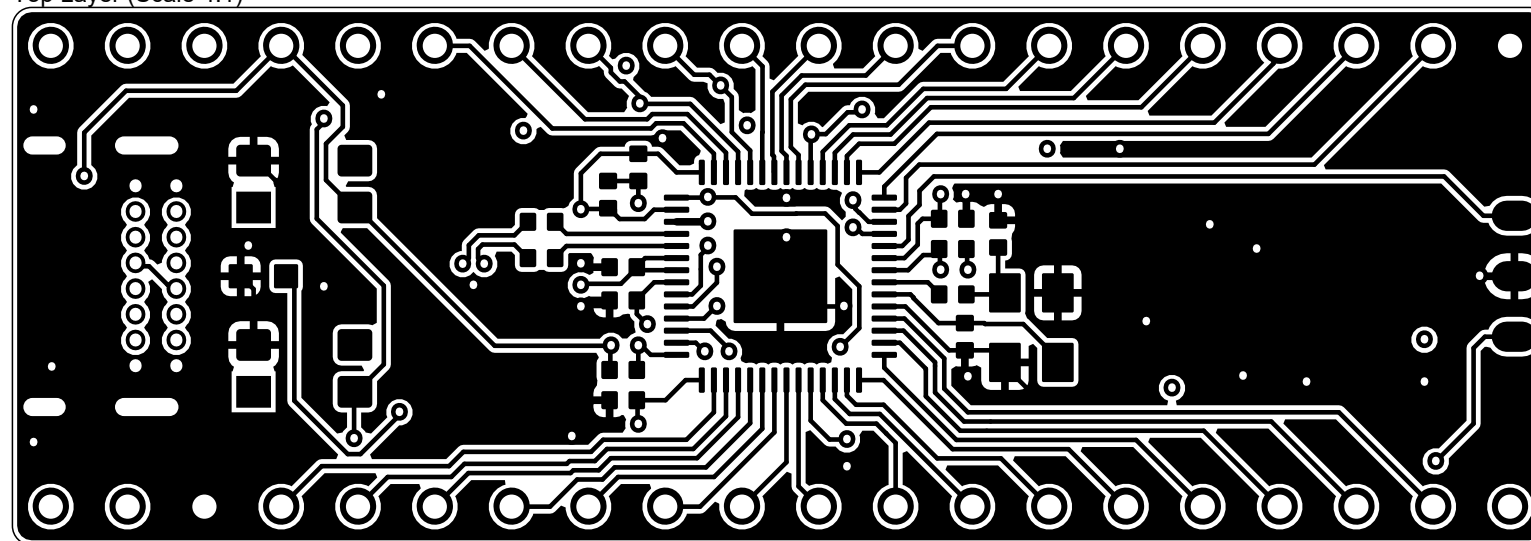
C

C

D

D

Top Layer (Scale 4:1)



LAYER VIEW : MID LAYER 1

A

A

B

B

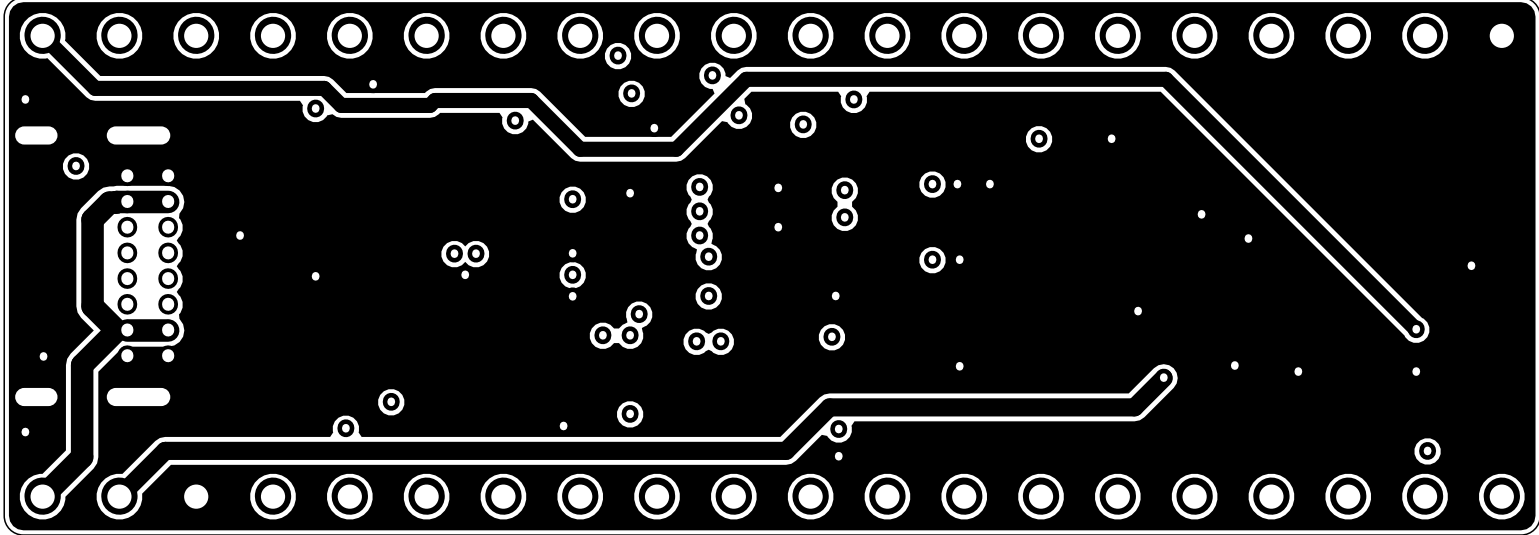
C

C

D

D

Layer 1 (Scale 4:1)



Project DIP2040.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : MID LAYER 2

A

A

B

B

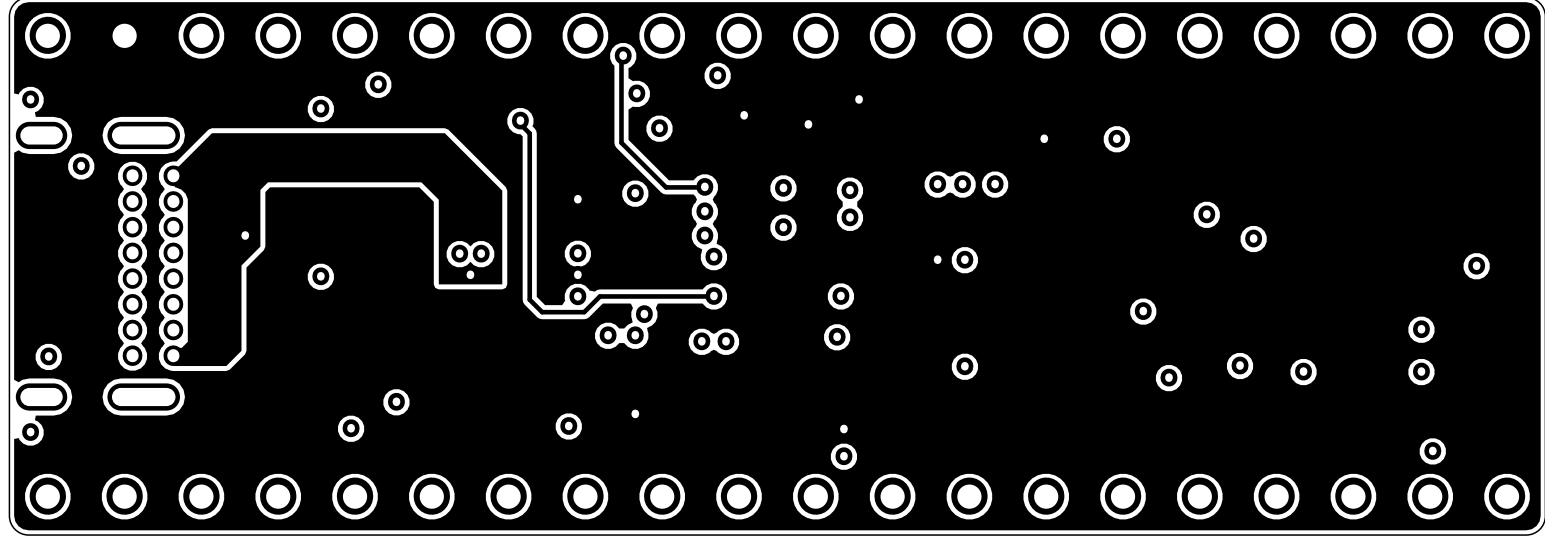
C

C

D

D

Layer 2 (Scale 4:1)

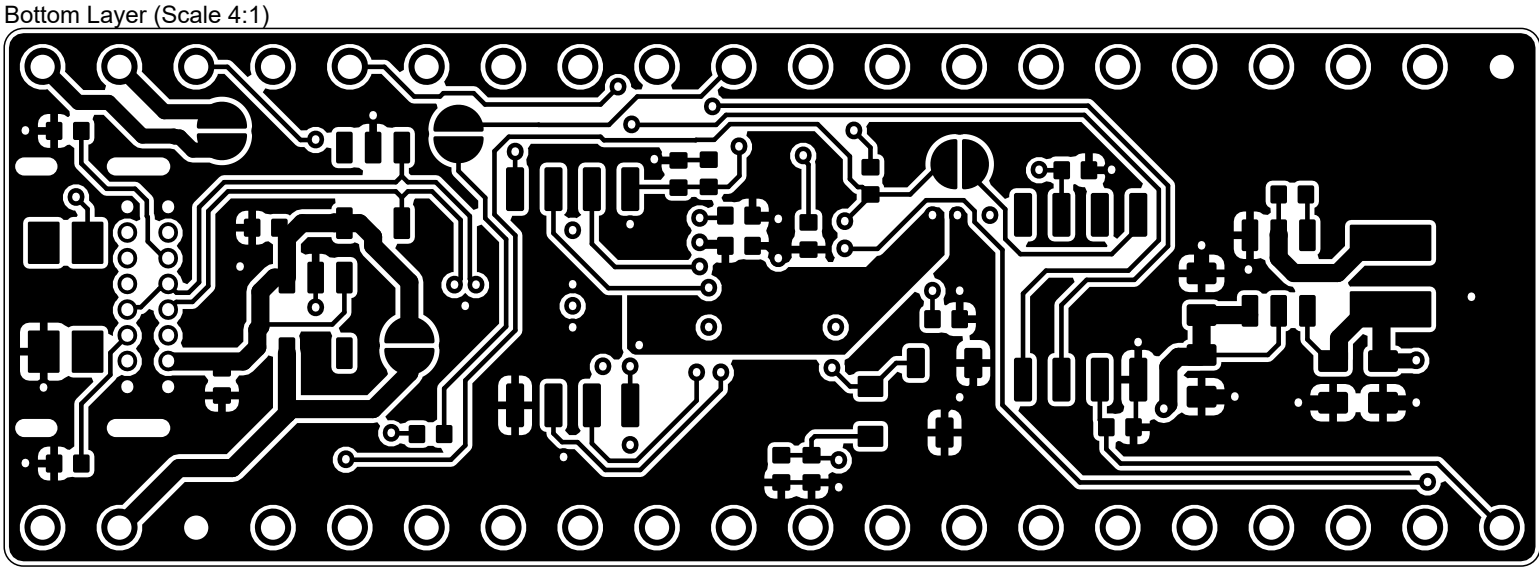


Project DIP2040.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : BOTTOM LAYER



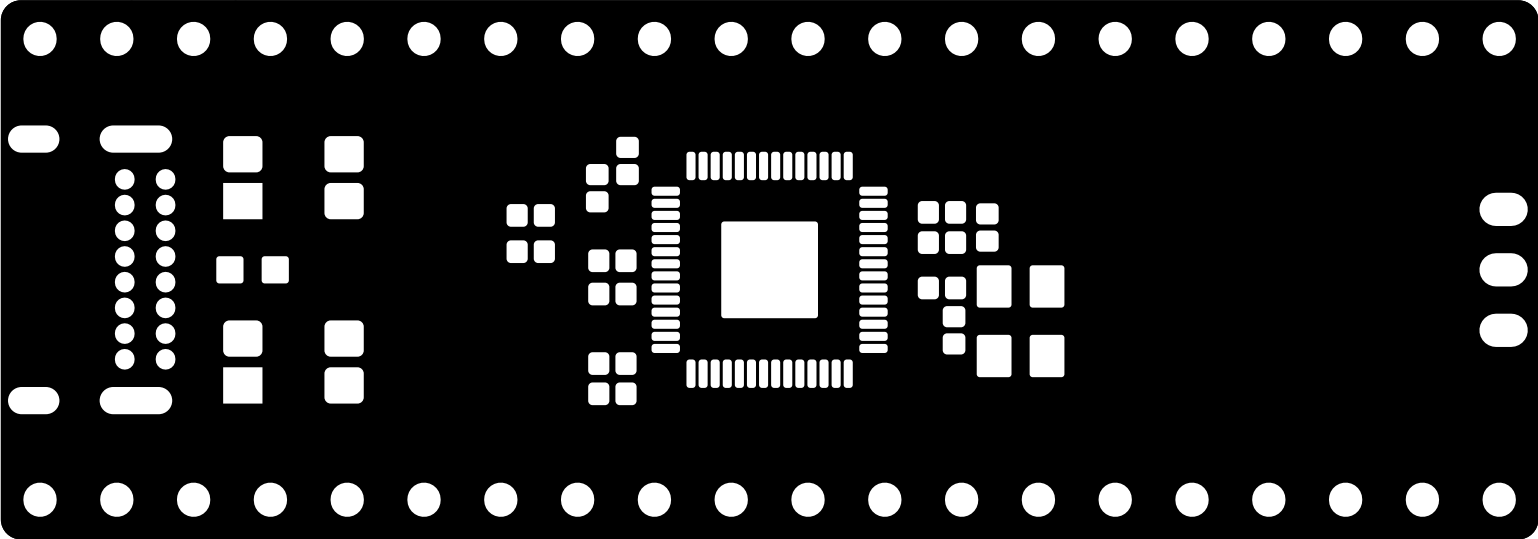
Project DIP2040.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)



67

68

69

70

71

72

LAYER VIEW : BOTTOM SOLDER MASK

A

A

B

B

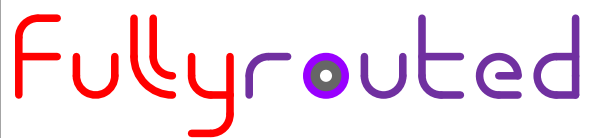
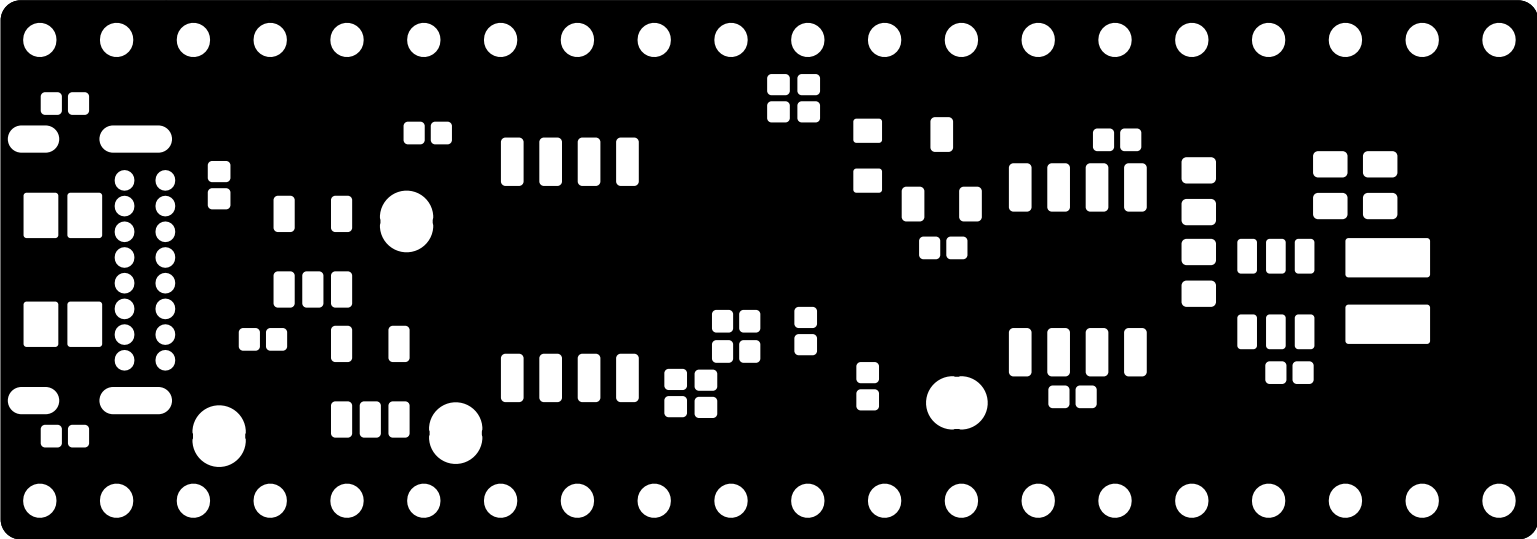
C

C

D

D

Bottom Solder (Scale 4:1)



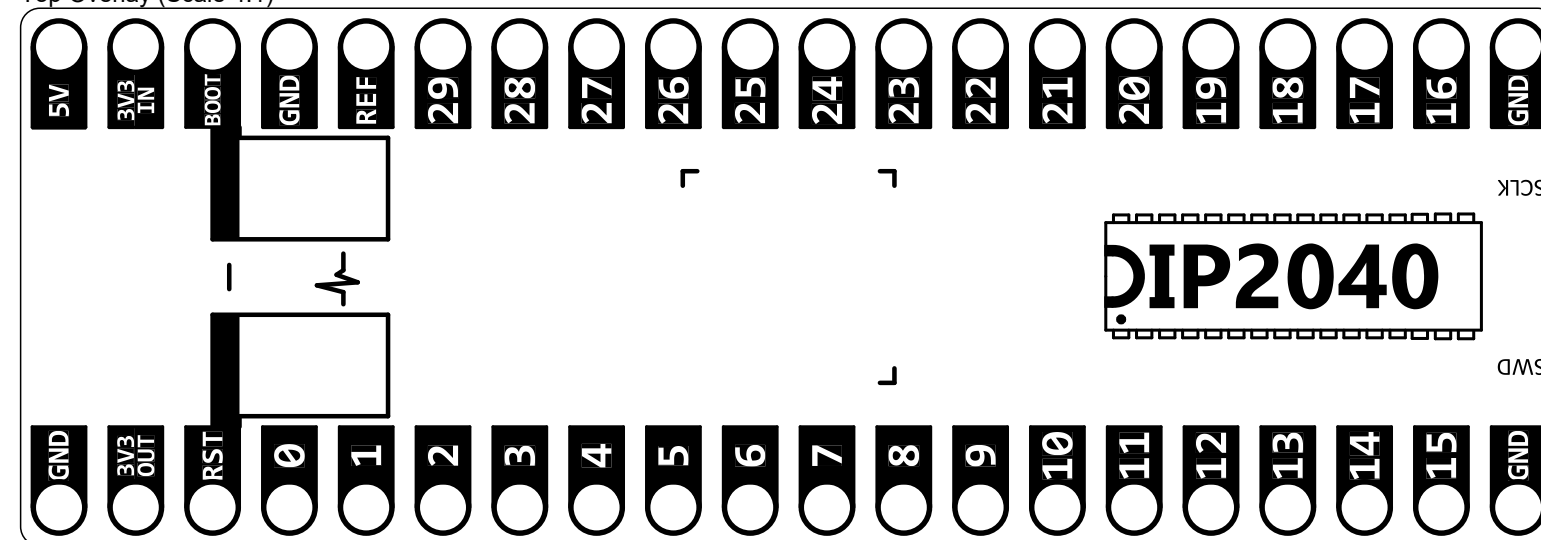
Project DIP2040.PrjPcb

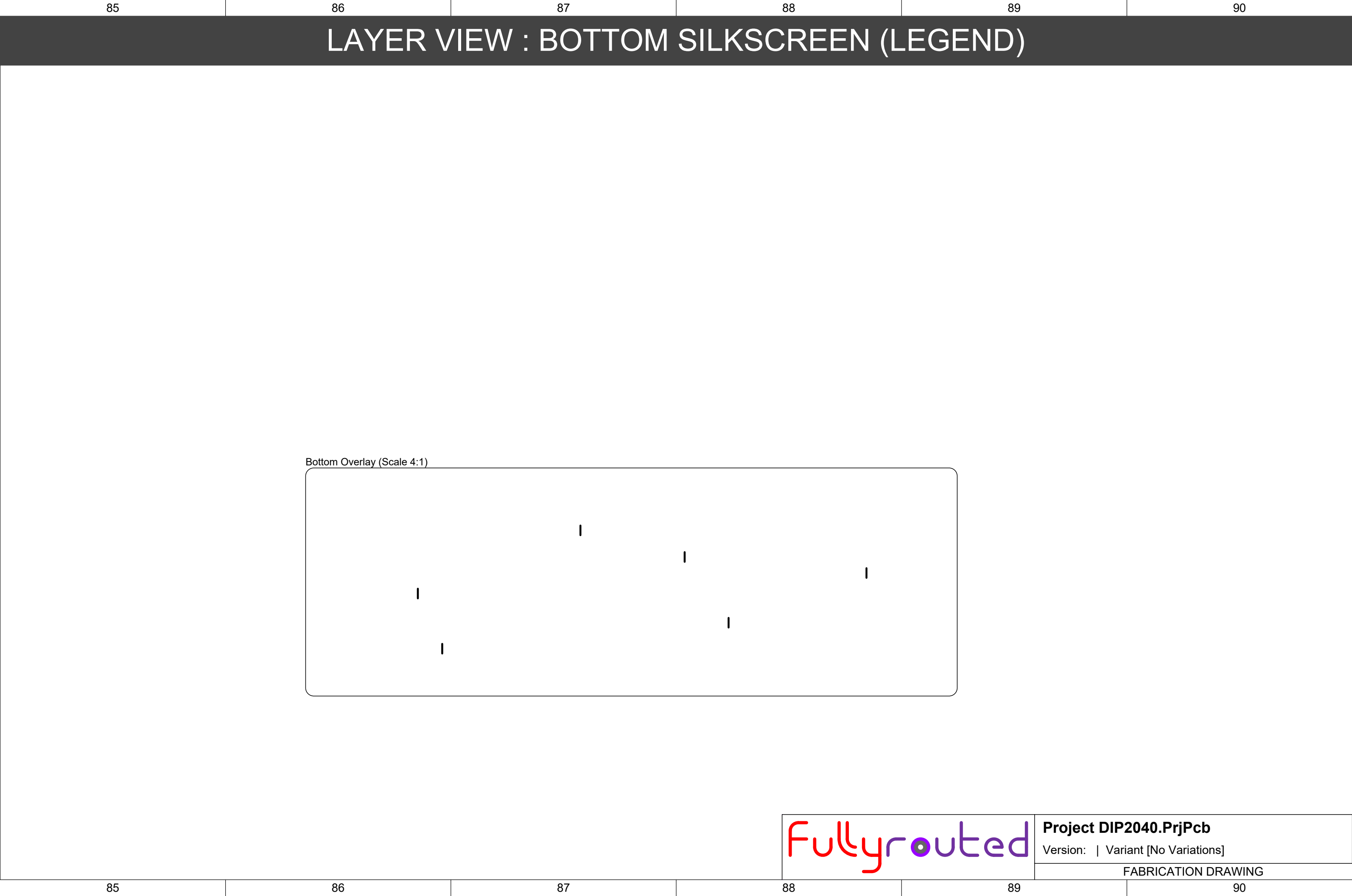
Version: | Variant [No Variations]

FABRICATION DRAWING

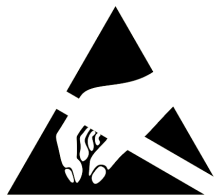
LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 4:1)





GENERAL



Unless otherwise specified the following rules apply:

- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

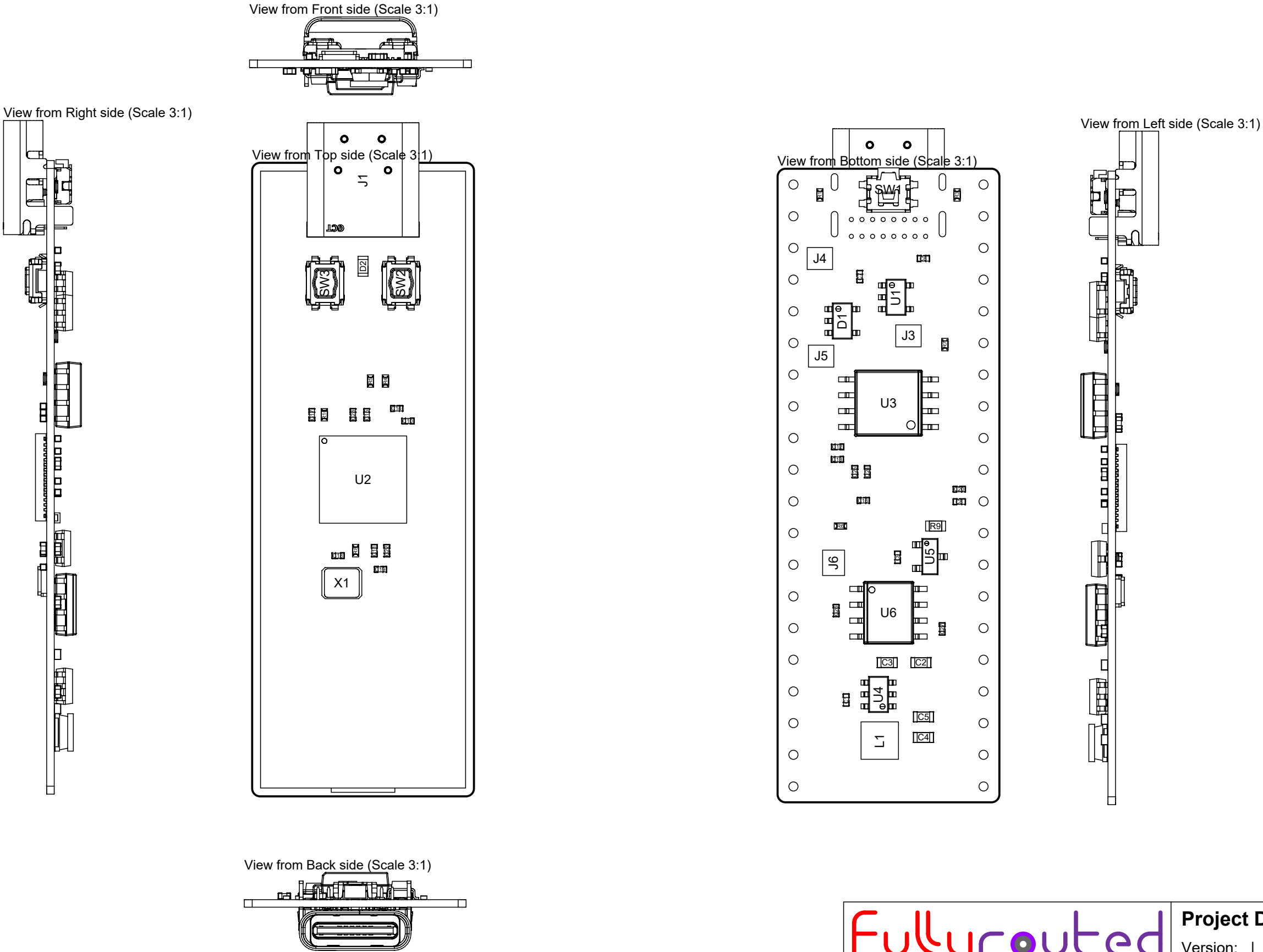
Bill of Materials and Material Handling

- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

Soldering

- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

2D VIEW



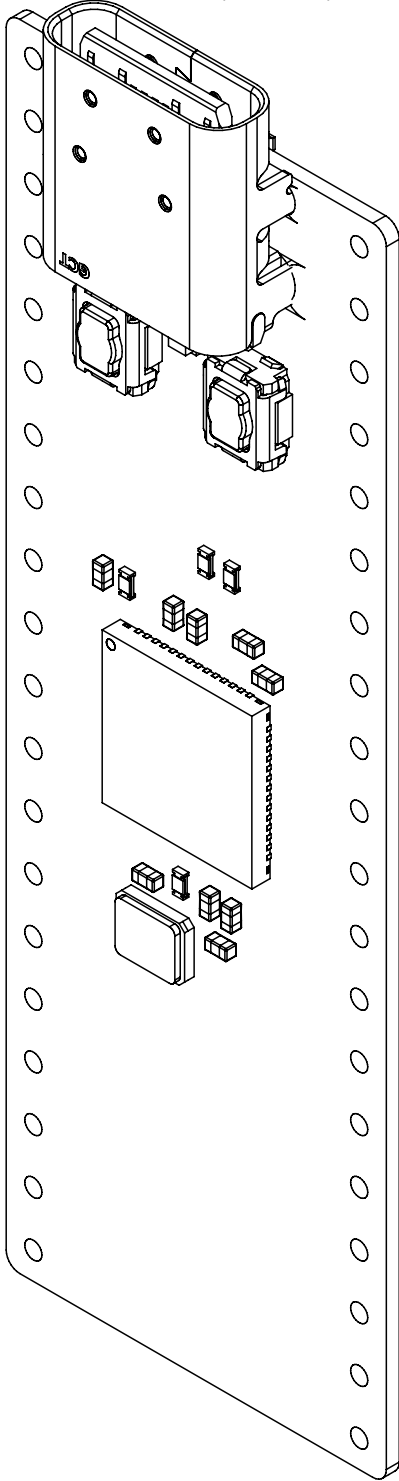
Project DIP2040.PrjPcb

Version: | Variant [No Variations]

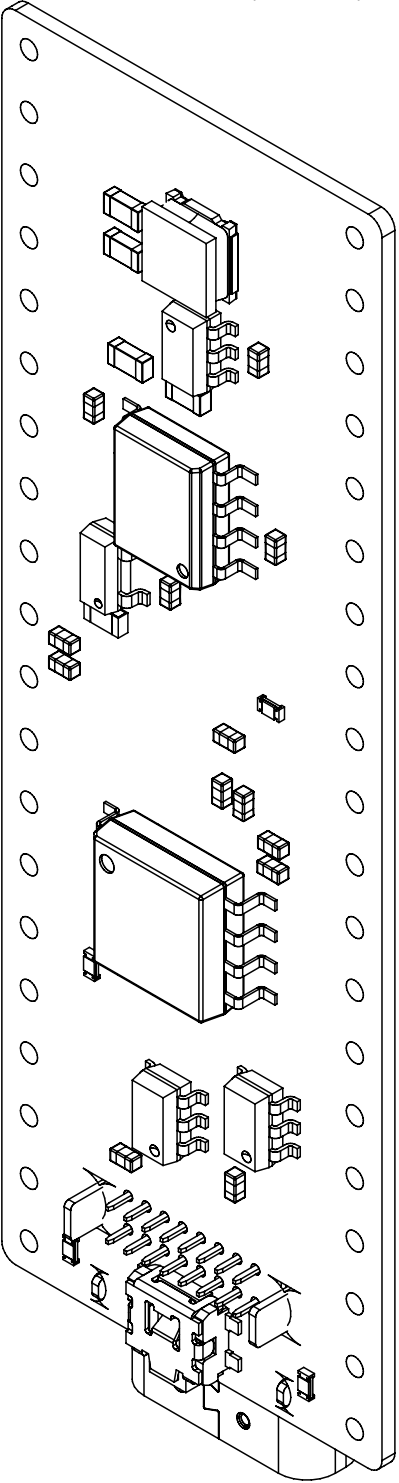
ASSEMBLY DRAWING

3D VIEW

View from Top side (Scale 4:1)



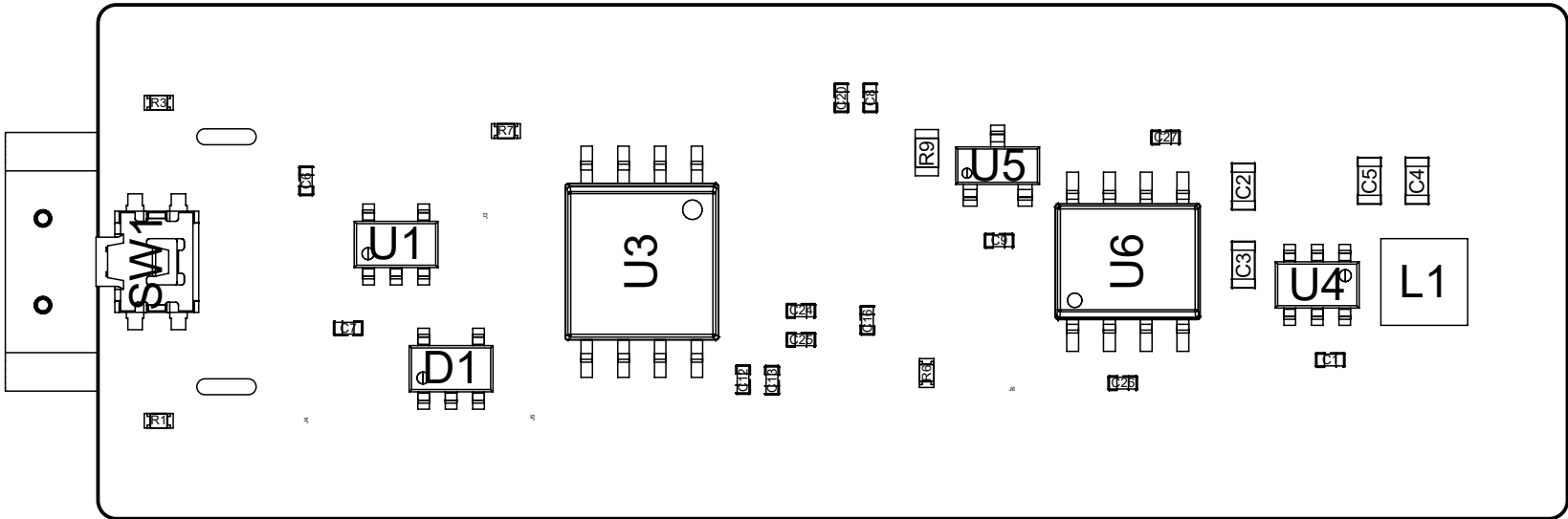
View from Bottom side (Scale 4:1)



1	2	3	4	5	6
Bill Of Materials					

DESIGNATORS BACK

View from Bottom side (Scale 4:1)



A

A

B

B

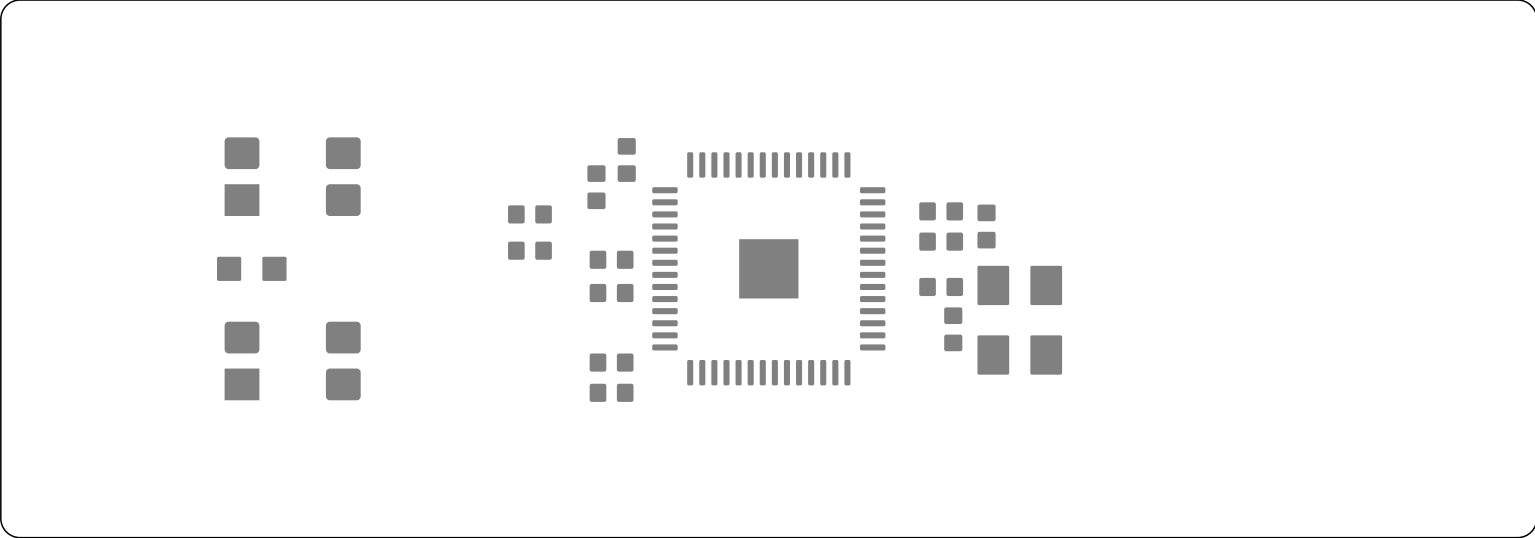
C

C

D

D

Top Paste (Scale 4:1)



PASTE MASK BACK

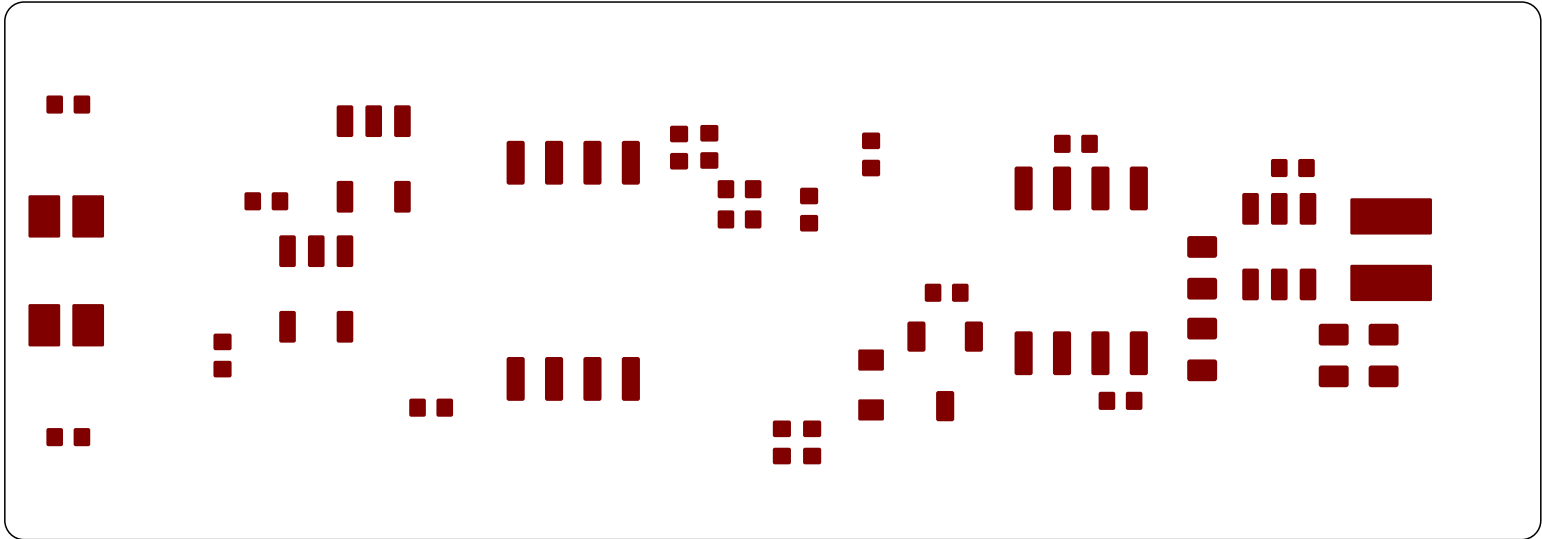
A

A

B

B

Bottom Paste (Scale 4:1)



C

C

D

D